

Exercise sheet 4

Topics in Topology, MTH602

1. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be covering maps. Show that the composition $g \circ f$ is a covering map if Z has a universal cover.
2. Prove that the fundamental group of a graph is always free. (*Hint*: Consider the maximal tree)
3. Prove that a cover of a graph is also a graph.
4. Use the previous two exercises to show that the algebraic fact that the subgroup of a free group is always free.
5. A cover whose deck transformation group is abelian is called an abelian cover. Prove that for a path connected, locally path connected, and semi-locally simply connected, there exists a “universal abelian cover”, i.e. an abelian cover that covers all other abelian covers.